CHAPTER 3

PARADOXES AND AXIOMS

In the preceding chapter we gave a brief exposition of the first, basic results of set theory, as it was created by Cantor and the pioneers who followed him in the last twenty five years of the 19th century. By the beginning of the 20th century, the theory had matured and justified itself with diverse and significant applications, especially in mathematical analysis. Perhaps its greatest success was the creation of an exceptionally beautiful and useful *transfinite arithmetic*, which introduces and studies the operations of addition, multiplication and exponentiation on infinite numbers. By 1900, there were still two fundamental problems about equinumerosity which remained unsolved. These have played a decisive role in the subsequent development of set theory and we will consider them carefully in the following chapters. Here we just state them, in the form of hypotheses.

3.1. Cardinal Comparability Hypothesis.[5] *For any two sets A, B, either $A \leq_c B$ or $B \leq_c A$.*

3.2. Continuum Hypothesis. *There is no set of real numbers X with cardinality intermediate between those of $\mathbb{N}$ and $\mathbb{R}$, i.e.,*

$$(\mathbf{CH}) \qquad (\forall X \subseteq \mathbb{R})[X \leq_c \mathbb{N} \vee X =_c \mathbb{R}].$$

Since $\mathbb{R} =_c \mathcal{P}(\mathbb{N})$, **CH** is a special case of the **Generalized Continuum Hypothesis**, the statement that for every infinite set A,

$$(\mathbf{GCH}) \qquad (\forall X \subseteq \mathcal{P}(A))[X \leq_c A \vee X =_c \mathcal{P}(A)].$$

If both of these hypotheses are true, then the natural numbers $\mathbb{N}$ and the reals $\mathbb{R}$ represent the two smallest "orders of infinity": every set is either countable, or equinumerous with $\mathbb{R}$, or strictly greater than $\mathbb{R}$ in cardinality.

In this beginning "naive" phase, set theory was developed on the basis of Cantor's definition of sets quoted in Chapter **1**, much as we proved its basic results in Chapter **2**. If we analyze carefully the proofs of those results, we

[5] Cantor announced the "theorem of comparability of cardinals" in 1895 and in 1899 he outlined a proposed proof of it in a letter to Dedekind, which was not, however, published until 1932. There were problems with that argument and it is probably closer to the truth to say that until 1900 (at least) the question of comparability of cardinals was still open.

will see that they are all based on the Extensionality Property (1-1) and the following simple assumption.

3.3. General Comprehension Principle. For each n-ary definite condition P, there is a set

$$A = \{\vec{x} \mid P(\vec{x})\}$$

whose members are precisely all the n-tuples of objects which satisfy $P(\vec{x})$, so that for all $\vec{x}$,

$$\vec{x} \in A \iff P(\vec{x}). \tag{3-1}$$

The extensionality principle implies that at most one set A can satisfy (3-1), and we call this A the **extension** of the condition P.

3.4. Definite conditions and operations. It is necessary to restrict the comprehension principle to definite conditions to avoid questions of vagueness which have nothing to do with science. We do not want to admit the "set"

$$A =_{\text{df}} \{x \mid x \text{ is an honest politician}\},$$

because membership of some specific public figure in it may be a hotly debated topic. *An n-ary* **condition** *P is* **definite** *if for each n-tuple $\vec{x} = (x_1, \dots, x_n)$ of objects, it is determined unambiguously whether $P(\vec{x})$ is true or false.* For example, the binary conditions P and S defined by

$$\begin{aligned} P(x,y) &\iff_{\text{df}} x \text{ is a parent of } y, \\ S(s,t) &\iff_{\text{df}} s \text{ and } t \text{ are siblings} \\ &\iff (\exists x)[P(x,s)\,\&\,P(x,t)] \end{aligned}$$

are both definite, assuming (for the example) that the laws of biology determine parenthood unambiguously. The General Comprehension Principle applies to them and we can form the sets of pairs

$$\begin{aligned} A &=_{\text{df}} \{(x,y) \mid x \text{ is a parent of } y\}, \\ B &=_{\text{df}} \{(s,t) \mid s \text{ and } t \text{ are siblings}\}. \end{aligned}$$

We do not demand of a definite condition that its truth value be *effectively determined.* For example, it is a famous open problem of number theory whether there exist infinitely many pairs of successive, odd primes, and the truth or falsity of the condition

$$G(n) \iff_{\text{df}} n \in \mathbb{N}\,\&\,(\exists m > n)[m, m+2 \text{ are both prime numbers}]$$

is not known for sufficiently large n. Still the condition G is unambiguous and we can use it to form the set of numbers

$$C =_{\text{df}} \{n \in \mathbb{N} \mid (\exists m > n)[m, m+2 \text{ are both prime numbers}]\}.$$

The *twin primes conjecture* asserts that $C = \mathbb{N}$, but if it is false, then C is some large, initial segment of the natural numbers.

In the same way, *an n-ary* **operation** *F is* **definite** *if it assigns to each n-tuple of objects $\vec{x}$ a unique, unambiguously determined object $w = F(\vec{x})$.* For example, assuming again that biology will not betray us, the operation

$$F(x) =_{\mathrm{df}} \begin{cases} \text{the father of } x, & \text{if } x \text{ is a human,} \\ x, & \text{otherwise,} \end{cases}$$

is definite. The silly consideration of cases here was put in to ensure that F determines a value for each argument x. In practice, we would define this operation by the simpler

$$F(x) =_{\mathrm{df}} \text{ the father of } x,$$

leaving it to the reader to supply some conventional, irrelevant value $F(x)$ for non-human x's. Again, definite operations need not be *effectively computable*, in fact the determination of the value $F(x)$ is sometimes the subject of judicial conflict in this specific case.

In addition to the General Comprehension Principle, we also assumed in the preceding chapter the existence of some specific sets, including the sets $\mathbb{N}$ and $\mathbb{R}$ of natural and real numbers, as well as the definiteness of some basic conditions from classical mathematics, e.g., the condition of "being a function",

$$\text{Function}(f, A, B) \iff f \text{ is a function from } A \text{ to } B.$$

This poses no problem as mathematicians have always made these assumptions, explicitly or implicitly.

The General Comprehension Principle has such strong intuitive appeal that the next theorem is called a "paradox".

3.5. Russell's paradox. *The General Comprehension Principle is not valid.*

PROOF. Notice first that if the General Comprehension Principle holds, then the set of all sets

$$V =_{\mathrm{df}} \{x \mid x \text{ is a set}\}$$

exists, and it has the peculiar property that it belongs to itself, $V \in V$. The common sets of everyday mathematics—sets of numbers, functions, etc.—surely do not belong to themselves, and so it is natural to consider them as members of a smaller, more natural universe of sets, by applying the General Comprehension Principle again,

$$R = \{x \mid x \text{ is a set and } x \notin x\}.$$

From the definition of R, however,

$$R \in R \iff R \notin R,$$

which is absurd. ⊣

When it is more than just a mistake, a "paradox" is simply a fact which runs counter to our intuitions, and set theorists already knew several such "paradoxes" before Russell announced this one in 1902, in a historic letter to the leading German philosopher and founder of mathematical logic Gottlob

Frege. These other paradoxes, however, were technical and affected only some of the most advanced parts of Cantor's theory. One could imagine that higher set theory had a systematic error built in, something like allowing a careless "division by 0" which would soon be discovered and disallowed, and then everything would be fixed. After all, contradictions and paradoxes had plagued the "infinitesimal calculus" of Newton and Leibnitz and they all went away after the rigorous foundation of the theory which was just being completed in the 1890s, without affecting the vital parts of the subject. Russell's paradox, however, was something else again: simple and brief, it affected directly the fundamental notion of set and the "obvious" principle of comprehension on which set theory had been built. It is not an exaggeration to say that Russell's paradox brought a foundational *crisis of doubt*, first to set theory and through it, later, to all of mathematics, which took over thirty years to overcome.

Some, like the French geometer Poincaré and the Dutch topologist and philosopher Brouwer, proposed radical solutions which essentially dismissed set theory (and much of classical mathematics along with it) as "pseudotheories", without objective content. From those who were reluctant to leave "Cantor's paradise", Russell first attempted to "rescue" set theory with his famous *theory of types*, which, however, is awkward to apply and was not accepted by a majority of mathematicians.[6] At approximately the same time, Zermelo proposed an alternative solution, which in time and with the contributions of many evolved into the contemporary theory of sets.

In his first publication on the subject in 1908, Zermelo took a pragmatic view of the problem. No doubt the General Comprehension Principle was not generally valid, Russell's paradox had made that clear. On the other hand, the specific applications of this principle in the proofs of basic facts about sets (like those in Chapter **2**) are few, simple and seemingly non-contradictory.

> Under such circumstances there is at this point nothing left for us to do but to proceed in the opposite direction [from that of the General Comprehension Principle] and, starting from set theory as it is historically given, to seek out the principles required for establishing the foundations of this mathematical discipline. In solving the problem we must, on the one hand, restrict these principles sufficiently to exclude all contradictions and, on the other, take them sufficiently wide to retain all that is valuable in this theory.

In other words, Zermelo proposed to replace the direct *intuitions* of Cantor about sets which led us to the faulty General Comprehension Principle with some *axioms*, hypotheses about sets which we accept with little a priori justification, simply because they are necessary for the proofs of the fundamental results of the existing theory and seemingly free of contradiction.

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[6]The theory of types has had a strong influence in the development of analytic philosophy and logic, and some of its basic ideas eventually have also found their place in set theory.

Such were the philosophically dubious beginnings of **axiomatic set theory**, surely one of the most significant achievements of 20th century science. From its inception, however, the new theory had a substantial advantage in the genius of Zermelo, who selected an extraordinarily natural and pliable axiomatic system. None of Zermelo's axioms has yet been discarded or seriously revised and (until very recently) only one basic new axiom was added to his seven in the decade 1920-1930. In addition, despite the opportunistic tone of the cited quotation, each of Zermelo's axioms expresses a property of sets which is intuitively obvious and was already well understood from its uses in classical mathematics. With the experience gained from working out the consequences of these axioms over the years, a new intuitive notion of "grounded set" has been created which does not lead to contradictions and for which the axioms of set theory are clearly true. We will reconsider the problem of foundation of set theory after we gain experience by the study of its basic mathematical results.

The basic model for the axiomatization of set theory was Euclidean geometry, which for 2000 years had been considered the "perfect" example of a rigorous, mathematical theory. If nothing else, the axiomatic method clears the waters and makes it possible to separate what might be confusing and self-contradictory in our intuitions about the objects we are studying, from simple errors in logic we might be making in our proofs. As we proceed in our study of axiomatic set theory, it will be useful to remind ourselves occasionally of the example of Euclidean geometry.

3.6. The axiomatic setup. We assume at the outset that there is a **domain** or **universe** $\mathcal{W}$ **of objects**, some of which are **sets**, and certain **definite conditions and operations** on $\mathcal{W}$, among them the basic conditions of *identity*, *sethood* and *membership*:

$$\begin{aligned} x = y &\iff x \text{ is the same object as } y, \\ \mathrm{Set}(x) &\iff x \text{ is a set}, \\ x \in y &\iff \mathrm{Set}(y) \text{ and } x \text{ is a member of } y. \end{aligned}$$

We call the objects in $\mathcal{W}$ which are not sets **atoms**, but we do not require that any atoms exist, i.e., we allow the possibility that all objects are sets. Definite conditions and operations are neither sets nor atoms.

This is the way every axiomatic theory begins. In Euclidean geometry for example, we start with the assumption that there are *points*, *lines* and several other geometrical objects and that some basic, definite conditions and operations are specified on them, e.g., it makes sense to ask if a "point P lies on the line L", or "to construct a line joining two given points". We then proceed to formulate the classical axioms of Euclid about these objects and to derive theorems from them. Actually Euclidean geometry is quite complex: there are several types of basic objects and a long list of intricate axioms about them. By contrast, Zermelo's set theory is quite austere: we just have sets and atoms and only seven fairly simple axioms relating them. In the remainder of this chapter we will introduce six of these axioms with a few comments and examples. It is a bit easier to put off stating his last, seventh axiom until we first gain some understanding of the consequences of the first six in the next few chapters.

3.7. (**I**) **Axiom of Extensionality.** *For any two sets A, B,*

$$A = B \iff (\forall x)[x \in A \iff x \in B].$$

3.8. (**II**) **Emptyset and Pairset Axiom**. (*a*) *There is a special object $\emptyset$ which we will call a set, but which has no members.* (*b*) *For any two objects x, y, there is a set A whose only members are x and y, so that it satisfies the equivalence*

$$t \in A \iff t = x \vee t = y. \tag{3-2}$$

The Axiom of Extensionality implies that only one empty set exists, and that for any two objects x, y, only one set A can satisfy (3-2). We denote this **doubleton** of x and y by

$$\{x, y\} =_{\text{df}} \text{ the unique set } A \text{ with sole members } x, y.$$

If $x = y$, then $\{x, x\} = \{x\}$ is the **singleton** of the object x.

Using this axiom we can construct many simple sets, e.g.,

$$\emptyset,\ \{\emptyset\},\ \{\{\emptyset\}\},\ \{\emptyset, \{\emptyset\}\},\ \{\{\emptyset\}, \{\{\emptyset\}\}\}, \dots,$$

but each of them has at most two members!

3.9. Exercise. *Prove that $\emptyset \neq \{\emptyset\}$.*

3.10. (**III**) **Separation Axiom** *or* **Axiom of Subsets**. *For each set A and each unary, definite condition P, there exists a set B which satisfies the equivalence*

$$x \in B \iff x \in A \,\&\, P(x). \tag{3-3}$$

From the Extensionality Axiom again, it follows that only one B can satisfy (3-3) and we will denote it by

$$B = \{x \in A \mid P(x)\}.$$

A characteristic contribution of Zermelo, this axiom is obviously a restriction of the General Comprehension Principle which implies many of its trouble-free consequences. For example, we can use it to define the operations of intersection and difference on sets,

$$A \cap B =_{\text{df}} \{x \in A \mid x \in B\},$$
$$A \setminus B =_{\text{df}} \{x \in A \mid x \notin B\}.$$

The proof of Russell's paradox yields a theorem:

3.11. Theorem. *For each set A, the set*

$$\boldsymbol{r}(A) =_{\text{df}} \{x \in A \mid x \notin x\} \tag{3-4}$$

is not a member of A. It follows that the collection of all sets is not a set, i.e., there is no set V such that

$$x \in V \iff \text{Set}(x).$$

PROOF. Notice first that $\boldsymbol{r}(A)$ is a set by the Separation Axiom. Assuming that $\boldsymbol{r}(A) \in A$, we have (as before) the equivalence

$$\boldsymbol{r}(A) \in \boldsymbol{r}(A) \iff \boldsymbol{r}(A) \notin \boldsymbol{r}(A),$$

which is absurd. $\dashv$

3.12. (**IV**) **Powerset Axiom**. *For each set A, there exists a set B whose members are the subsets of A, i.e.,*

$$X \in B \iff \mathrm{Set}(X) \,\&\, X \subseteq A. \tag{3-5}$$

Here $X \subseteq A$ is an abbreviation of $(\forall t)[t \in X \Longrightarrow t \in A]$. The Axiom of Extensionality implies that for each A, only one set B can satisfy (3-5); we call it the **powerset** of A and we denote it by

$$\mathcal{P}(A) =_{\mathrm{df}} \{X \mid \mathrm{Set}(X) \,\&\, X \subseteq A\}.$$

3.13. Exercise. $\mathcal{P}(\emptyset) = \{\emptyset\}$ *and* $\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$.

3.14. Exercise. *For each set A, there exists a set B whose members are exactly all singletons of members of A, i.e.,*

$$x \in B \iff (\exists t \in A)[x = \{t\}].$$

3.15. (**V**) **Unionset Axiom.** *For each set $\mathscr{E}$, there exists a set B whose members are the members of the members of $\mathscr{E}$, i.e., so that it satisfies the equivalence*

$$t \in B \iff (\exists X \in \mathscr{E})[t \in X]. \tag{3-6}$$

The Axiom of Extensionality implies again that for each $\mathscr{E}$, only one set can satisfy (3-6); we call it the **unionset** of $\mathscr{E}$ and we denote it by

$$\bigcup \mathscr{E} =_{\mathrm{df}} \{t \mid (\exists X \in \mathscr{E})[t \in X]\}.$$

The unionset operation is obviously most useful when $\mathscr{E}$ is a **family of sets**, i.e., a set all of whose members are also sets. This is the case for the simplest application, which (finally) gives us the binary, union operation on sets: we set

$$A \cup B = \bigcup \{A, B\}$$

using axioms (**II**) and (**V**), and we compute

$$\begin{aligned} t \in A \cup B &\iff (\exists X \in \{A, B\})[t \in X] \\ &\iff t \in A \vee t \in B. \end{aligned}$$

3.16. Exercise. $\bigcup \emptyset = \bigcup \{\emptyset\} = \emptyset$.

3.17. (**VI**) **Axiom of Infinity.** *There exists a set I which contains the empty set $\emptyset$ and the singleton of each of its members, i.e.,*

$$\emptyset \in I \,\&\, (\forall x)[x \in I \Longrightarrow \{x\} \in I].$$

We have not given yet a rigorous definition of "infinite", but it is quite obvious that any I with the properties in the axiom must be infinite, since (**VI**) implies

$$\emptyset \in I,\ \{\emptyset\} \in I,\ \{\{\emptyset\}\} \in I, \ldots$$

and the objects $\emptyset$, $\{\emptyset\}$, ... are all distinct sets by the Extensionality Axiom. The intuitive understanding of the axiom is that it demands precisely the existence of the set

$$I = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \dots\},$$

but it is simpler (and sufficient) to assume of I only the stated properties, which imply that it contains all these complex singletons.

It was a commonplace belief among philosophers and mathematicians of the 19th century that the existence of infinite sets could be proved, and in particular the set of natural numbers could be "constructed" out of thin air, "by logic alone". All the proposed "proofs" involved the faulty General Comprehension Principle in some form or another. We know better now: logic can codify the valid forms of reasoning but it cannot prove the existence of anything, let alone infinite sets. By taking account of this fact cleanly and explicitly in the formulation of his axioms, Zermelo made a substantial contribution to the process of purging logic of ontological concerns and separating the mathematical development of the theory of sets from logic, to the benefit of both disciplines.

3.18. Axioms for definite conditions and operations. Zermelo understood *definite conditions* intuitively, he described them much as we did in **3.4** and he applied the Separation Axiom using various quite complex conditions without any special argument that they are, indeed, "definite". We will do the same, because the business of proving definiteness is boring and not particularly illuminating. For the sake of completeness, however, we list here the only properties of definiteness that we will actually use.

1. The following basic conditions are definite:

$$\begin{aligned} x = y &\iff_{\mathrm{df}} x \text{ and } y \text{ are the same object}, \\ \mathrm{Set}(x) &\iff_{\mathrm{df}} x \text{ is a set}, \\ x \in y &\iff_{\mathrm{df}} \mathrm{Set}(y) \text{ and } x \text{ is a member of } y, \end{aligned}$$

2. For each object c and each n, the constant n-ary operation

$$F(x_1, \dots, x_n) = c$$

is definite.

3. Each *projection operation*

$$F_i(x_1, \dots, x_n) = x_i \quad (1 \le i \le n)$$

is definite.

4. If P is a definite condition of $n + 1$ arguments and for each tuple of objects $\vec{x} = x_1, \dots, x_n$ there exists exactly one w such that $P(\vec{x}, w)$ is true, then the operation

$$F(\vec{x}) = \text{the unique } w \text{ such that } P(\vec{x}, w)$$

is definite.

5. If Q is an m-ary definite condition, each F_i is an n-ary definite operation for $i = 1, \dots, m$ and
$$P(\vec{x}) \iff_{\text{df}} Q(F_1(\vec{x}), \dots, F_m(\vec{x})),$$
then the condition P is also definite.
6. If Q, R and S are definite conditions of the appropriate number of arguments, then so are the following conditions which are obtained from them by applying the elementary operations of logic:
$$\begin{aligned}
P_1(\vec{x}) &\iff_{\text{df}} \neg P(\vec{x}) && \iff && P(\vec{x}) \text{ is false,}\\
P_2(\vec{x}) &\iff_{\text{df}} Q(\vec{x})\,\&\,R(\vec{x}) && \iff && \text{both } Q(\vec{x}) \text{ and } R(\vec{x}) \text{ are true,}\\
P_3(\vec{x}) &\iff_{\text{df}} Q(\vec{x}) \vee R(\vec{x}) && \iff && \text{either } Q(\vec{x}) \text{ or } R(\vec{x}) \text{ is true,}\\
P_4(\vec{x}) &\iff_{\text{df}} (\exists y)S(\vec{x}, y) && \iff && \text{for some } y, S(\vec{x}, y) \text{ is true,}\\
P_5(\vec{x}) &\iff_{\text{df}} (\forall y)S(\vec{x}, y) && \iff && \text{for every } y, S(\vec{x}, y) \text{ is true.}
\end{aligned}$$

All the conditions and operations we will use can be proved definite by appealing to these basic properties. Aside from one problem at the end of this chapter, however, for the logically minded, we will omit these technical proofs of definiteness and it is best for the reader to forget about them too: they detract from the business at hand, which is the study of sets, not definite conditions and operations.

3.19. Classes. Having gone to all the trouble to discredit the General Principle of Comprehension, we will now profess that *for every unary, definite condition P there exists a* **class**
$$A = \{x \mid P(x)\}, \tag{3-7}$$
such that for every object x,
$$x \in A \iff P(x). \tag{3-8}$$

To give meaning to this principle and prove it, we need to define classes. Every set will be a class, but because of the Russell Paradox **3.5**, there must be classes which are not sets, else (3-8) leads immediately to the Russell Paradox in the case $P(x) \iff \text{Set}(x)\,\&\,x \notin x$.

First let us agree that for every unary, definite condition P we will write synonymously
$$x \in P \iff P(x).$$
For example, if Set is the basic condition of sethood, we write interchangeably
$$x \in \text{Set} \iff \text{Set}(x) \iff x \text{ is a set.}$$
This is just a convenient notational convention.

A unary definite condition P is **coextensive** with a set A if the objects which satisfy it are precisely the members of A, in symbols
$$P =_{\text{e}} A \iff_{\text{df}} (\forall x)[P(x) \iff x \in A]. \tag{3-9}$$
For example, if
$$P(x) \iff x \neq x,$$

then $P =_{\mathrm{e}} \emptyset$. By the Russell Paradox **3.5**, not every P is coextensive with a set. On the other hand, *a unary, definite condition P is coextensive with at most one set*; because if $P =_{\mathrm{e}} A$ and also $P =_{\mathrm{e}} B$, then for every x,

$$x \in A \iff P(x) \iff x \in B,$$

and $A = B$ by the Axiom of Extensionality.

By definition, a **class** *is either a set or a unary definite condition which is not coextensive with a set*. With each unary condition P, we associate the class

$$\{x \mid P(x)\} =_{\mathrm{df}} \begin{cases} \text{the unique set } A \text{ such that } P =_{\mathrm{e}} A, \\ \qquad\qquad\qquad\qquad \text{if } P =_{\mathrm{e}} A \text{ for some set } A, \\ P, \quad \text{otherwise.} \end{cases} \tag{3-10}$$

Now if $A =_{\mathrm{df}} \{x \mid P(x)\}$, then either P is coextensive with a set, in which case $P =_{\mathrm{e}} A$ and by the definition $x \in A \iff P(x)$; or P is not coextensive with any set, in which case $A = P$ and

$$x \in A \iff x \in P \iff P(x) \quad \text{(by the notational convention).}$$

This is exactly the *General Comprehension Principle for Classes* enunciated in (3-7) and (3-8).

3.20. Exercise. *For every set A,*

$$\{x \mid x \in A\} = A,$$

and, in particular, every set is a class. Show also that

$$\{X \mid \mathrm{Set}(X) \,\&\, X \subseteq A\} = \mathcal{P}(A).$$

3.21. Exercise. *The class $\mathcal{W}$ of all objects is not a set, and neither is the class* Set *of all sets.*

If P is an n-ary definite condition and F an n-ary definite operation, we set

$$\{F(\vec{x}) \mid P(\vec{x})\} =_{\mathrm{df}} \{w \mid (\exists\vec{x})[P(\vec{x}) \,\&\, w = F(\vec{x})]\}. \tag{3-11}$$

For example, with $F(x) = \{x\}$,

$$\{\{x\} \mid x = x\} = \{w \mid (\exists x)[w = \{x\}]\} = \text{the class of all singletons.}$$

3.22. Exercise. *The class $\{\{x\} \mid x = x\}$ of all singletons is not a set.*

3.23. Exercise. *For every class A,*

$$\begin{aligned} A \text{ is a set} &\iff \text{for some class } B,\ A \in B \\ &\iff \text{for some set } X,\ A \subseteq X, \end{aligned}$$

where inclusion among classes is defined as if they were sets,

$$A \subseteq B \iff_{\mathrm{df}} (\forall x)[x \in A \implies x \in B].$$

3.24. The Axioms of Choice and Replacement: a warning. Our axiomatization of set theory will not be complete until we introduce Zermelo's last Axiom of Choice in Chapter **8** and the later Axiom of Replacement in Chapter **11**. While there are good reasons for these postponements which we will explain in due course, there are also good reasons for adding the axioms of Choice and Replacement: many basic set theoretic arguments need them, and among these are some of the simplest claims of Chapter **2**. Thus, until Chapter **8**, we will need to be extra careful and make sure that our constructions indeed can be justified by axioms (**I**) – (**VI**) and that we have not sneaked in some "obvious" assertion about sets not yet proved or assumed. In a few places we will formulate and prove something weaker than the whole truth, when the proof of the whole truth needs one of the missing axioms. Now this is good: it will keep us on our toes and make us understand better the art of reasoning from axioms.

3.25. About atoms. Most recent developments of axiomatic set theory assume at the outset the so-called **Principle of Purity**, that *there are no atoms*: all objects of the basic domain are *sets*. There is a certain appealing simplicity to this conception of a mathematical world in which everything is a set. We have followed Zermelo in allowing atoms (without demanding them), primarily because this makes the theory more naturally relevant to the natural sciences: we want our results to apply to sets of *planets*, *molecules* or *frogs*, and frogs are not sets. In any case, it comes at little cost, we simply have to say "object" in some situations where the atom banners would say "set". It is important to notice, however, that none of the axioms requires the existence of atoms, so none of the consequences we will derive from them depends on the existence of atoms: everything we will prove remains true in the domain of pure sets, provided only that it satisfies the Zermelo axioms, as we stated them.

3.26. Axioms as closure properties of the universe $\mathcal{W}$. Whatever the domain $\mathcal{W}$ of our axiomatic set theory may be, it is clear that it does not contain all "objects of our intuition or thought" in Cantor's expression; $\mathcal{W}$ is not a set, and it is certainly a perfectly legitimate mathematical object of our intuition about which we intend to have many thoughts. Granting that $\mathcal{W}$ is not all there is, we can fruitfully conceive of the axioms as imposing *closure conditions* on it. We have assumed (so far) that $\mathcal{W}$ contains $\emptyset$, that it is closed under the operations of pairing $\{x, y\}$ (**II**), powerset $\mathcal{P}(X)$ (**IV**), and unionset $\bigcup\mathcal{E}$ (**V**), that it includes every definite subcollection of every set (**III**), and that it contains some set I with the stipulated property of the Axiom of Infinity (**VI**). It is also possible to understand the Axiom of Extensionality (**I**) as a closure property of $\mathcal{W}$: in its non-trivial direction, it says that for any two sets A, B,

$$A \neq B \Longrightarrow (\exists t)[t \in (A \setminus B) \cup (B \setminus A)], \tag{3-12}$$

i.e., every inequality $A \neq B$ between two sets is witnessed by some legitimate object $t \in \mathcal{W}$ which belongs to one and not to the other.

This understanding of the meaning of the axioms is compatible with two different conceptions of the universe $\mathcal{W}$. One is that it is huge, amorphous, difficult to understand and impossible to define; but every object in it is concrete, definite, whole, and this is enough to justify the closure properties of $\mathcal{W}$ embodied by the axioms. Let us call this *the large view*. The *small view* is that $\mathcal{W}$ consists precisely of those objects whose existence is "guaranteed" by the axioms, those which can be "constructed" by applying the axioms repeatedly: the axioms are satisfied because we deliberately put in $\mathcal{W}$ all the objects required by the closure properties they express. Neither conception is precise, to be sure, but they are different. On the small view, for example, there are no atoms since none of the axioms demands their existence, while the large view allows frogs among the objects in $\mathcal{W}$.

Both of these views can be defended and they have played significant roles in the philosophy of set theory, and even in its mathematical practice, by suggesting the kind of questions one should ask. We will come back to discuss the issue in Chapter **11** and Appendix **B**, when we will be in a position to be less flippant about it. In the meantime, we will often speak of the axioms as closure conditions on $\mathcal{W}$, a useful heuristic device which is compatible with every philosophical approach to the subject.

Problems for Chapter 3

x3.1. For each non-empty set $\mathscr{E}$ and each $X \in \mathscr{E}$, we define the *intersection of $\mathscr{E}$ via X* by

$$\bigcap{}_X \mathscr{E} =_{\text{df}} \{x \in X \mid (\forall U \in \mathscr{E})[x \in U]\}.$$

Show that for any two members X, Y of $\mathscr{E}$,

$$\bigcap{}_X \mathscr{E} = \bigcap{}_Y \mathscr{E},$$

i.e., the intersection $\bigcap_X \mathscr{E}$ is independent of the specific X we used in its definition, and hence we can use for it the notation $\bigcap \mathscr{E}$ which does not exhibit X. Show also that $A \cap B = \bigcap\{A, B\}$.

x3.2. For any two sets A, B, determine whether each of the following classes is or is not a set.

1. $\{\{\emptyset, x\} \mid x \in A\}$.
2. $\{x \mid \text{Set}(x) \,\&\, x \neq \emptyset\}$.
3. $\{\{x, y\} \mid x \in A \,\&\, y \in B\}$.
4. $\{\mathcal{P}(X) \mid X \subseteq A\}$.

x3.3. Prove rigorously that the following conditions and operations are definite, using only (**I**) – (**VI**) and the axioms in **3.18**. (Here c is some arbitrary object.)

$$\begin{aligned}
P_1(x) &\Longleftrightarrow_{\text{df}} x \in c, & F_1(x, y) &=_{\text{df}} \{x, y\},\\
P_2(x, y, z) &\Longleftrightarrow_{\text{df}} z \in x, & F_2(X) &=_{\text{df}} \bigcup X,\\
P_3(X, Y) &\Longleftrightarrow_{\text{df}} X \subseteq Y, & F_3(X) &=_{\text{df}} \mathcal{P}(X),\\
P_4(X, Y) &\Longleftrightarrow_{\text{df}} X \cap Y = \emptyset, & F_4(x) &=_{\text{df}} \{x\},\\
P_5(X, Y) &\Longleftrightarrow_{\text{df}} \mathcal{P}(X) \subseteq Y. & F_5(X, Y) &=_{\text{df}} X \cup Y.
\end{aligned}$$

In the remaining problems of this section we consider Zermelo's notion of *equivalence*, which intuitively holds between two sets when they are equinumerous. These problems will be trivial after we introduce *functions* within

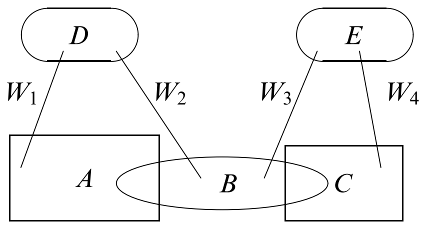

FIGURE **3.1**. The hypothesis for Part 3 of Problem **x3.5**.

Zermelo's axiomatic theory in the next Chapter, but right now they are challenging.

3.27. Definition. Recall that two sets A, B are **disjoint** if $A \cap B = \emptyset$. A set W is a **connection** of the two disjoint sets A and B (according to Zermelo) if the following three conditions hold:

1. $Z \in W \Longrightarrow (\exists x \in A, y \in B)[Z = \{x, y\}]$.
2. For each $x \in A$, there is exactly one $y \in B$ such that $\{x, y\} \in W$.
3. For each $y \in B$, there is exactly one $x \in A$ such that $\{x, y\} \in W$.

x3.4. For any two disjoint sets A, B, the class

$$\Sigma(A, B) = \{W \mid W \text{ is a connection of } A \text{ with } B\}$$

is a set.

3.28. Definition. Two sets A, B are **equivalent according to Zermelo** if there exists a third set C disjoint from both of them and connections of A with C and of B with C, in symbols

$$\begin{aligned} A \sim_Z B \iff & (\exists C, W, W')[A \cap C = \emptyset \,\&\, B \cap C = \emptyset \\ & \qquad \& \, W \in \Sigma(A, C) \,\&\, W' \in \Sigma(B, C)]. \end{aligned}$$

*__x3.5.__ The condition of equivalence according to Zermelo has the following properties, for any three sets A, B, C:

$$A \sim_Z A,$$

$$\text{if } A \sim_Z B, \text{ then } B \sim_Z A,$$

$$\text{if } (A \sim_Z B \,\&\, B \sim_Z C), \text{ then } A \sim_Z C.$$

HINT. To prove $A \sim_Z A$, you need to find some set C such that $A \cap C = \emptyset$ and there is a connection W of A with C. The hypothesis for the (last) transitivity property is illustrated in Figure **3.1**.